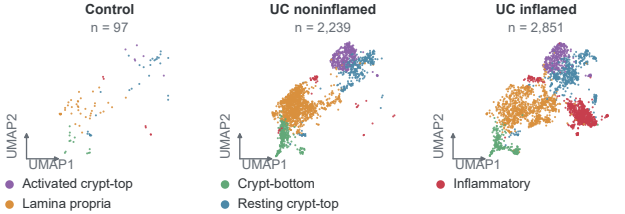


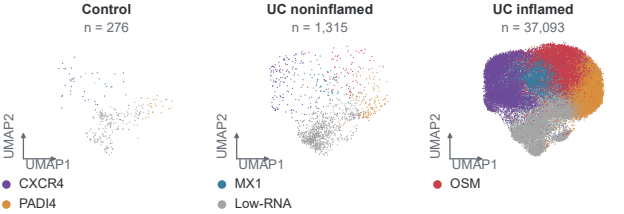
Figure 1

a Human UC fibroblast and neutrophil state landscape

Fibroblasts



Neutrophils



b Paired cell-state remodeling

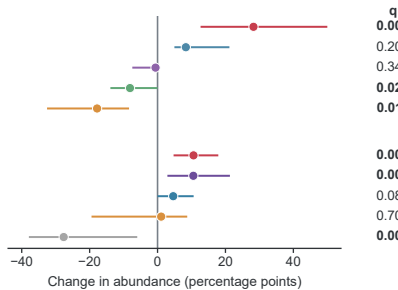
Inflamed minus noninflamed; median and 95% CI

Fibroblasts | 13 patient pairs

Inflammatory
Resting crypt-top
Activated crypt-top
Crypt-bottom
Lamina propria

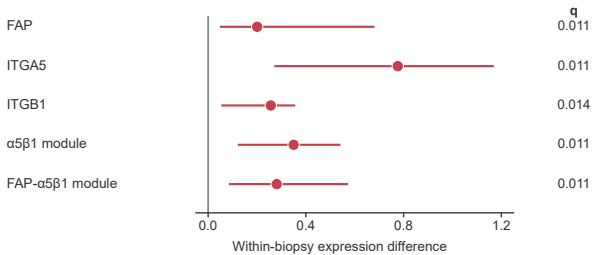
Neutrophils | 17 patient pairs

OSM
CXCR4
MX1
PADI4
Low-RNA



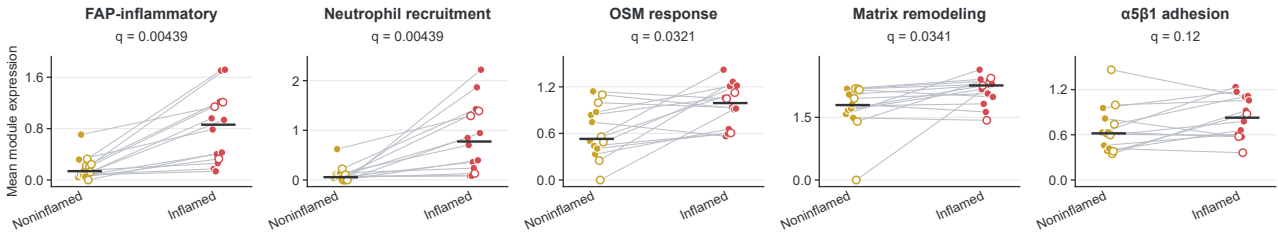
c Inflammatory-fibroblast activation

Inflammatory vs other fibroblasts; 10 biopsies



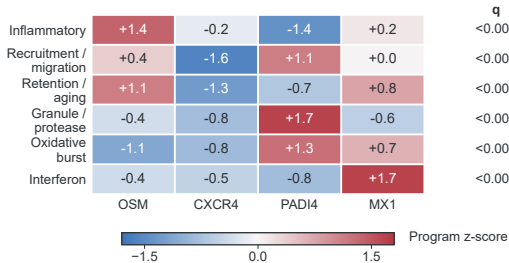
d Paired fibroblast program changes

14 patient pairs; lines connect patients; black bars, medians; open points, <10 recovered cells



e Neutrophil state-associated RNA programs

11 matched patients; ≥20 cells/state; naming genes excluded



Equal patient weights; Friedman q (BH, 9 programs).

f Predicted reciprocal signals

Fibroblast → neutrophil

HGF → CD44
CXCL1 → CXCR2
CSF3 → CSF3R

Neutrophil → fibroblast

OSM → OSMR
IL1B → IL1R1
NAMPT → ITGA5
NAMPT → ITGB1

LR-expressing fraction

Paired coexpression: Extended Data Fig. 1c.

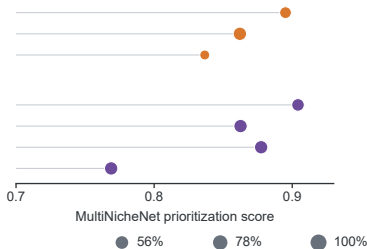


Figure 2

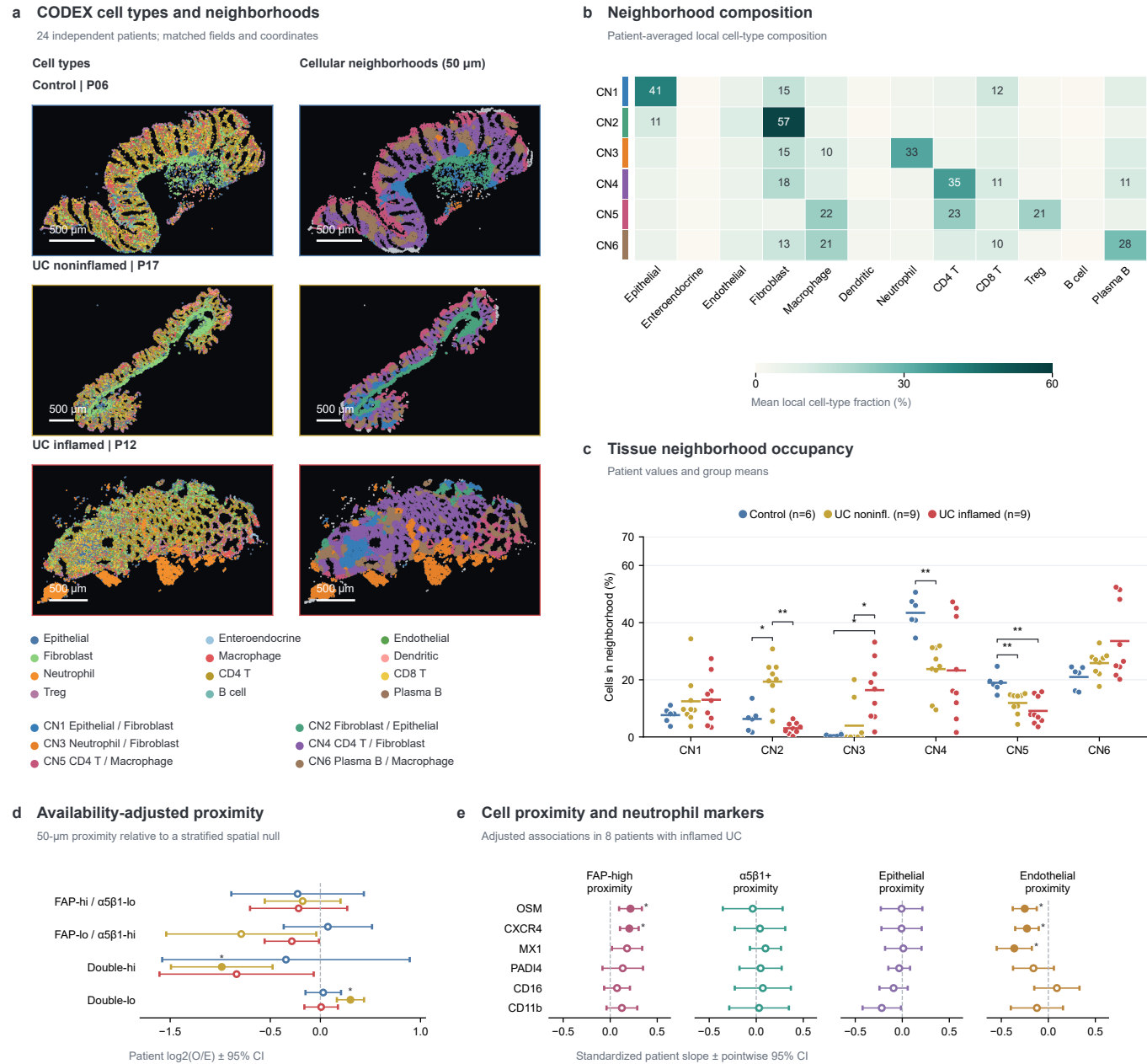
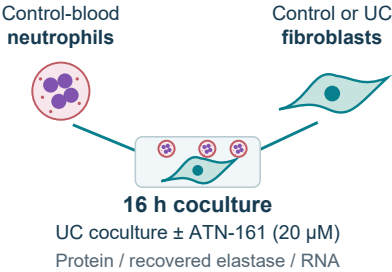
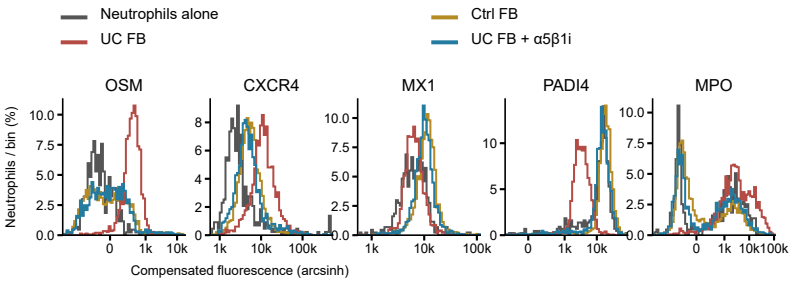


Figure 3

a Human coculture workflow



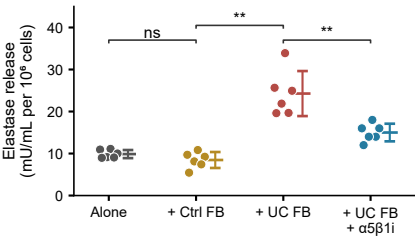
b Neutrophil protein distributions



c Fibroblast exposure and blockade effects on neutrophil proteins

	Fibroblast exposure			Neutrophils alone		Blockade in UC FB coculture				Δ protein z score
	Ctrl FB vs alone	UC FB vs alone	UC FB vs Ctrl FB	NAMPTi vs untreated	α5β1i vs untreated	NAMPTi vs untreated	α5β1i vs untreated	Dual vs untreated	Dual vs NAMPTi	
OSM	+0.07	+1.87**	+1.80**	+0.03	-0.19	-0.01	-1.96**	-1.85**	-1.84*	
CXCR4	+0.83**	+2.86**	+2.03**	-0.04	+0.65**	-1.09*	-2.14**	-2.16**	-1.07**	
MX1	+1.43*	-0.44	-1.87**	+0.47	+1.24*	+0.54	+1.97**	+1.54**	+1.01*	
PADI4	+0.37	-1.75**	-2.11**	+0.13	+0.42	+0.12	+2.32**	+2.07**	+1.95**	
MPO	-0.48*	+2.07*	+2.56**	-0.19	-0.57*	-1.27	-2.13*	-2.18**	-0.91*	

d NETosis Assay (Elastase Release)



e Neutrophil RNA response programs

	Ctrl FB vs alone	UC FB vs alone	UC vs Ctrl FB	α5β1i alone	α5β1i in UC FB	Δ protein z score
Inflammatory cytokines	+0.32*	+0.40***	+0.08	+0.35***	-0.07	
Interferon response	+0.06*	+0.04	-0.01	+0.06*	+0.01	
Granule/protease	+0.02	+0.02	-0.02	+0.07	+0.06***	
NADPH oxidase	+0.25**	+0.28**	-0.02	+0.28***	-0.05	
Chemotaxis	+0.46***	+0.48***	-0.00	+0.50***	-0.14*	
Hypoxia/glycolysis	+0.19**	+0.14***	-0.06	+0.20***	+0.01	

f Predicted fibroblast–neutrophil routes

	UC vs Ctrl FB	α5β1i vs UC FB
1 COL1A2 → CD44 ECM-FB → OSM	-0.79	+0.82
2 FN1 → CD44 ECM-FB → CXCR4	-0.64	+0.67
3 IL1B → MX1/ISG CCN-FB → MX1/ISG	-0.45	+0.62
4 CEACAM1 → CEACAM8 ECM-FB → PADI4	+0.41	-0.41
5 CXCL12 → CXCR4 CCN-FB → CXCR4	+0.65	-0.65
6 C5 → C5AR1 ECM-FB → CXCR4	+0.79	-0.79

LR consensus change (−0.9 to +0.9)

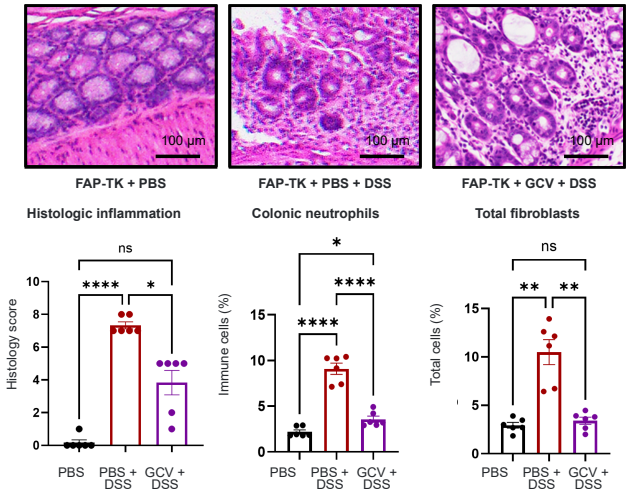
g Module support for ligands in f

	Inflam. cytokines	IFN response	Granule/protease	NADPH oxidase	Chemo-taxis	Hypoxia/glycolysis	Δ protein z score
1 COL1A2	5	6	2	1	1	8	
2 FN1	5	10	2	1	1	7	
3 IL1B	8	10	3	0	4	2	
4 CEACAM1	5	10	2	2	2	7	
5 CXCL12	7	10	2	2	1	7	
6 C5	8	10	2	1	1	7	

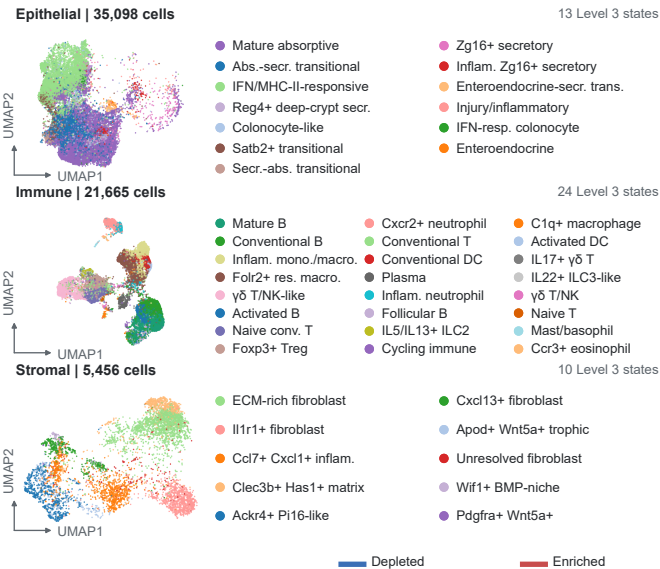
Matched ligands; color: prior rank; dot size/count: top decile

Figure 4

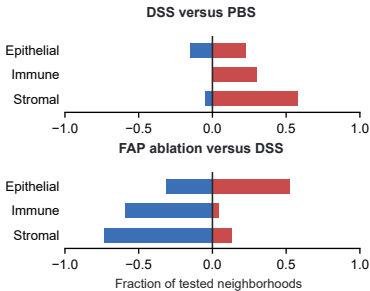
a FAP ablation reduces DSS-colitis pathology



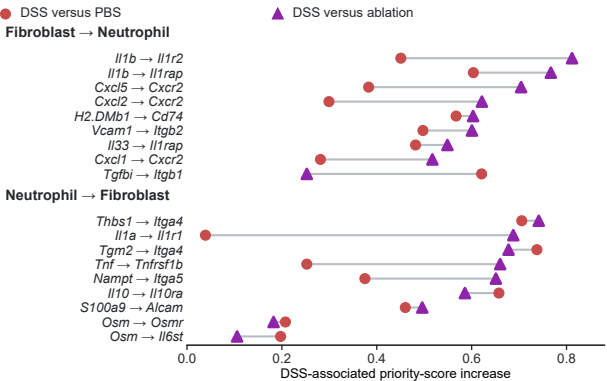
b Compartment-specific UMAPs



c MitoR neighborhood remodeling



d Reciprocal signalling routes



e Predicted receiver-cell pathways

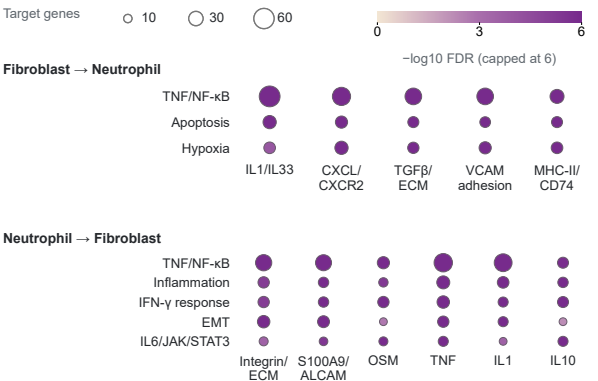
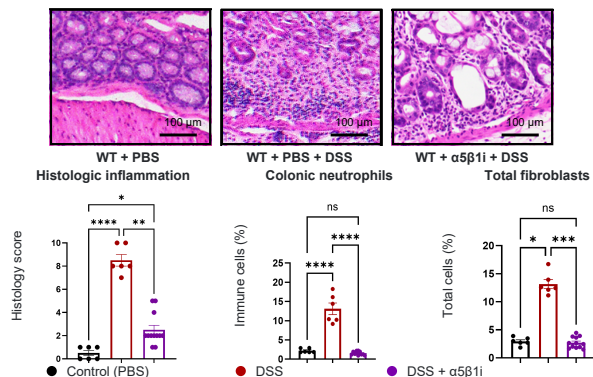
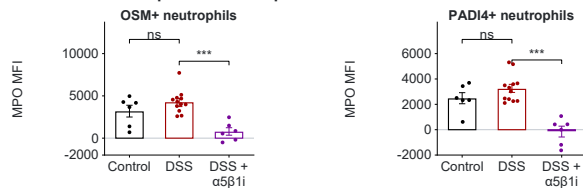


Figure 5

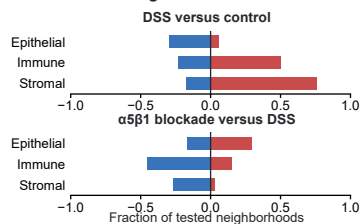
a $\alpha 5\beta 1$ blockade reduces DSS-colitis pathology



b MPO in marker-positive neutrophils

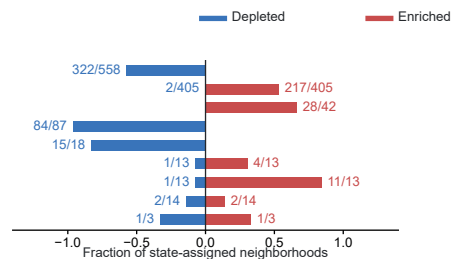


c Milor neighborhood remodeling

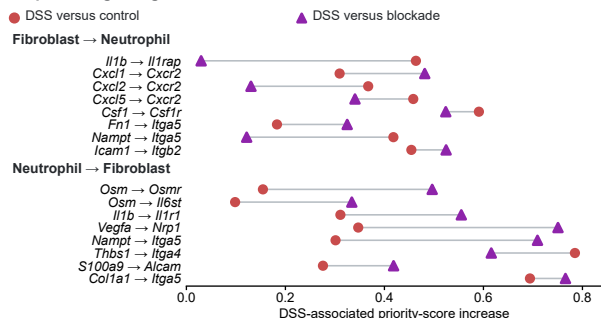


Level 3 states | blockade versus DSS

IFN/MHC-II-responsive
Injury/inflammatory epithelial
Inflam. Zg16+ secretory epithelial
Conventional B cell
IL1r1+ fibroblast
Osm-associated neutrophil
Pad14-associated neutrophil
Mx1-associated neutrophil
Cxcr4-associated neutrophil



d Reciprocal signaling routes



e Predicted receiver-cell pathways

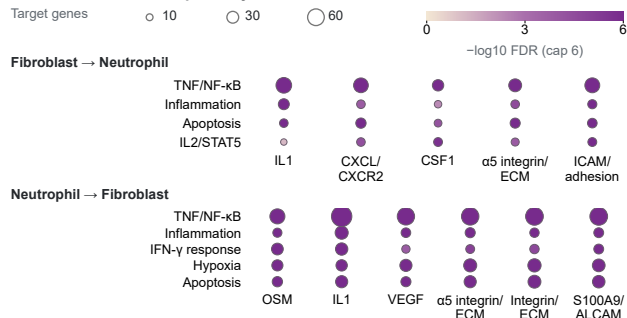
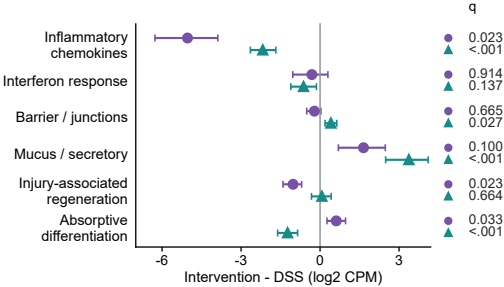


Figure 6

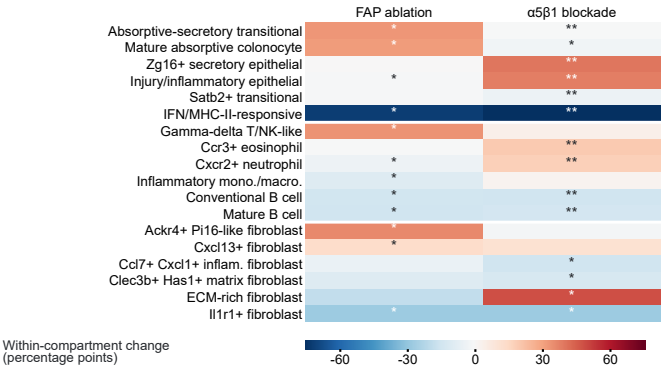
a Epithelial RNA programs

Whole epithelium | DSS n=6; ablation n=4; blockade n=12

FAP ablation α5β1 blockade

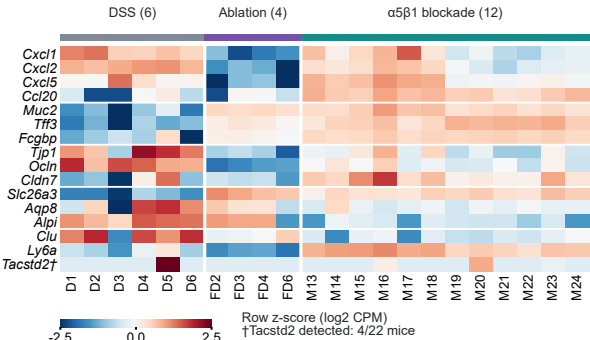


c Cell-state composition



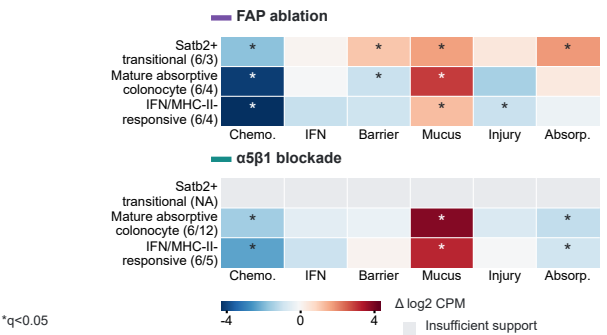
e Epithelial gene expression

Mature absorptive colonocytes | 16 selected genes



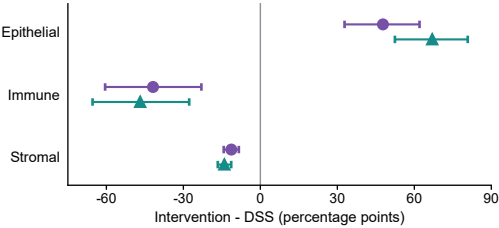
b Within-state epithelial programs

≥20 cells per mouse; n = DSS / intervention



d Recovered-cell proportions

DSS n=6; ablation n=5; blockade n=12



Shared DSS reference; separate interventions.

RNA programs do not measure epithelial repair.

f Individual-mouse epithelial profiles

Whole epithelium | n=6 / 4 / 12 mice

DSS FAP ablation α5β1 blockade

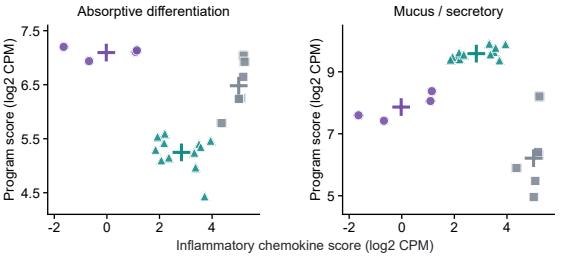
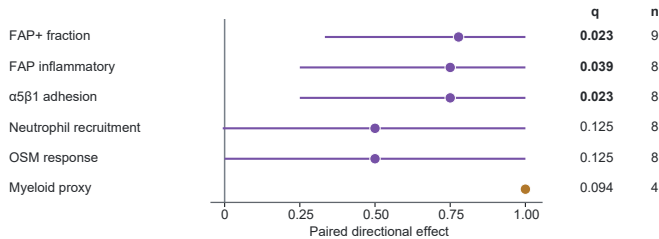


Figure 7

a TAURUS UC: paired inflammation

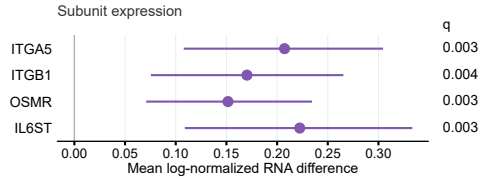
Inflamed vs noninflamed; 95% bootstrap CI



Myeloid: monocyte proxy; TAURUS did not retain neutrophils.

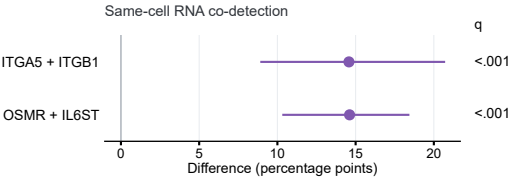
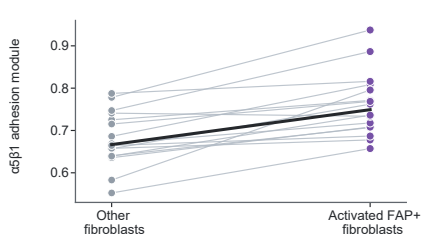
c Receptor availability in activated FAP+ fibroblasts

Activated FAP+ minus other fibroblasts; 16 paired patients; 50 matched biopsies



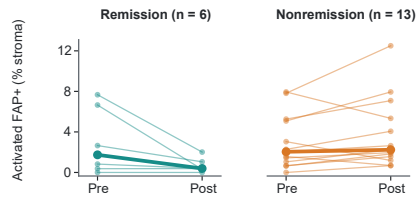
b FAP+ fibroblast adhesion state

16 patients; median paired Δ = 0.07



d Activated FAP+ cells after anti-TNF

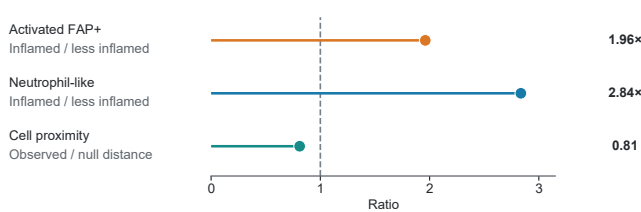
Site-matched pre/post adalimumab



Lines: patients; bold lines: group medians

e Paired spatial validation in UC

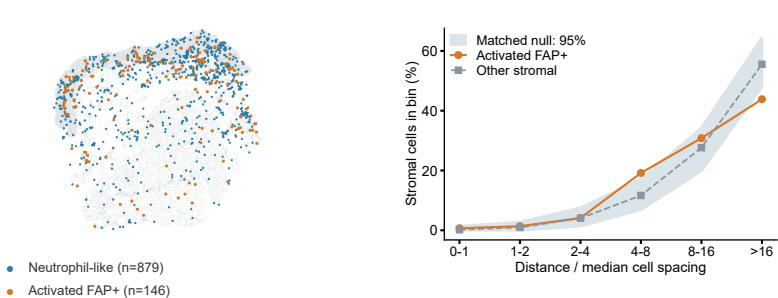
One donor; paired UC sections (SCP3818)



Proximity: P = 0.005; 1,000 size-matched null sets

f Spatial context of the activated-FAP niche

UC1 inflamed; one donor



g Cross-system pathway responses

